Chapter 2

Solution of Algebraic and Transcendental Equations

2.1 INTRODUCTION

In scientific and engineering studies, a frequently occurring problem is to find the roots of equations of the form

$$f(x) = 0 \tag{2.1}$$

If $f(x)$ is a quadratic, cubic or a biquadratic expression, then algebraic formulae are available for expressing the roots in terms of the coefficients. On the other hand, when $f(x)$ is a polynomial of higher degree or an expression involving transcendental functions, algebraic methods are not available, and recourse must be taken to find the roots by approximate methods.

This chapter is concerned with the description of several numerical methods for the solution of equations of the form given in Eq. (2.1), where $f(x)$ is algebraic or transcendental or a combination of both. Now, algebraic functions of the form

$$f_n(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n, \tag{2.2}$$

are called *polynomials* and we discuss some special methods for determining their roots. A non-algebraic function is called a *transcendental* function, e.g., $f(x) = \ln x^3 - 0.7$ $\phi(x) = e^{-0.5x} - 5x$, $\psi(x) = \sin^2 x - x^2 - 2$, etc. The roots of Eq. (2.1) may be either real or complex. We discuss methods of finding a real root of algebraic or transcendental equations and also methods of determining all real and complex roots of polynomials. Solution of systems of nonlinear equations will be considered at the end of the chapter.

If $f(x)$ is a polynomial of the form Eq. (2.2), the following results, from the theory of equations would be useful in locating its roots.

(i) Every polynomial equation of the nth degree has n and only n roots.

(ii) If n is odd, the polynomial equation has atleast one real root whose sign is opposite to that of the last term.

(iii) If n is even and the constant term is negative, then the equation has atleast one positive root and atleast one negative root.

(iv) If the polynomial equation has (a) real coefficients, then imaginary roots occur in pairs and (b) rational coefficients, then irrational roots occur in pairs.

(v) *Descartes' Rule of Signs*

(a) A polynomial equation $f(x) = 0$ cannot have more number of positive real roots than the number of changes of sign in the coefficients of $f(x)$.

(b) In (a) above, $f(x) = 0$ cannot have more number of negative real roots than the number of changes of sign in the coefficients of $f(-x)$.

2.2 BISECTION METHOD

This method is based on Theorem 1.1 which states that if a function $f(x)$ is continuous between a and b, and $f(a)$ and $f(b)$ are of opposite signs, then there exists at least one root between a and b. For definiteness, let $f(a)$ be negative and $f(b)$ be positive. Then the root lies between a and b and let its approximate value be given by $x_0 = (a + b)/2$. If $f(x_0) = 0$, we conclude that x_0 is a root of the equation $f(x) = 0$. Otherwise, the root lies either between x_0 and b, or between x_0 and a depending on whether $f(x_0)$ is negative or positive. We designate this new interval as $[a_1, b_1]$ whose length is $|b - a|/2$. As before, this is bisected at x_1 and the new interval will be exactly half the length of the previous one. We repeat this process until the latest interval (which contains the root) is as small as desired, say ε. It is clear that the interval width is reduced by a factor of one-half at each step and at the end of the nth step, the new interval will be $[a_n, b_n]$ of length $|b - a|/2^n$. We then have

$$\frac{|b-a|}{2^n} \leq \varepsilon,$$

which gives on simplification

$$n \geq \frac{\log_e(|b-a|/\varepsilon)}{\log_e 2} \tag{2.3}$$

Equation (2.3) gives the number of iterations required to achieve an accuracy ε. For example, if $|b - a| = 1$ and $\varepsilon = 0.001$, then it can be seen that

$$n \geq 10 \tag{2.4}$$

The method is shown graphically in Fig. 2.1.

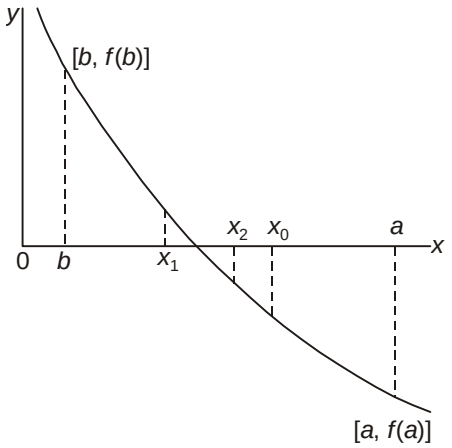


Figure 2.1 Graphical representation of the bisection method.

It should be noted that this method always succeeds. If there are more roots than one in the interval, bisection method finds one of the roots. It can be easily programmed using the following computational steps:

1. Choose two real numbers a and b such that $f(a)\, f(b) < 0$.
2. Set $x_r = (a + b)/2$.
3. (a) If $f(a)\, f(x_r) < 0$, the root lies in the interval (a, x_r). Then, set $b = x_r$ and go to step 2 above.
   (b) If $f(a)\, f(x_r) > 0$, the root lies in the interval (x_r, b). Then, set $a = x_r$ and go to step 2.
   (c) If $f(a)\, f(x_r) = 0$, it means that x_r is a root of the equation $f(x) = 0$ and the computation may be terminated.

In practical problems, the roots may not be exact so that condition (c) above is never satisfied. In such a case, we need to adopt a criterion for deciding when to terminate the computations.

A convenient criterion is to compute the percentage error ε_r defined by

$$\varepsilon_r = \left| \frac{x_r' - x_r}{x_r'} \right| \times 100\%. \tag{2.5}$$

where x_r' is the new value of x_r. The computations can be terminated when ε_r becomes less than a prescribed tolerance, say ε_p. In addition, the maximum number of iterations may also be specified in advance.

Example 2.1 Find a real root of the equation $f(x) = x^3 - x - 1 = 0$.

Since $f(1)$ is negative and $f(2)$ positive, a root lies between 1 and 2 and, therefore, we take $x_0 = 3/2$. Then

$$f(x_0) = \frac{27}{8} - \frac{3}{2} = \frac{15}{8}, \text{ which is positive.}$$

Hence the root lies between 1 and 1.5 and we obtain

$$x_1 = \frac{1+1.5}{2} = 1.25$$

We find $f(x_1) = -19/64$, which is negative. We, therefore, conclude that the root lies between 1.25 and 1.5. If follows that

$$x_2 = \frac{1.25+1.5}{2} = 1.375$$

The procedure is repeated and the successive approximations are

$$x_3 = 1.3125, \quad x_4 = 1.34375, \quad x_5 = 1.328125, \text{ etc.}$$

Example 2.2 Find a real root of the equation $x^3 - 2x - 5 = 0$.

Let $f(x) = x^3 - 2x - 5$. Then

$$f(2) = -1 \quad \text{and} \quad f(3) = 16.$$

Hence a root lies between 2 and 3 and we take

$$x_1 = \frac{2+3}{2} = 2.5$$

Since $f(x_1) = f(2.5) = 5.6250$, the root lies between 2 and 2.25.
Hence

$$x_2 = \frac{2+2.5}{2} = 2.25$$

Now, $f(x_2) = 1.890625$, the root lies between 2 and 2.25.
Therefore,

$$x_3 = \frac{2+2.25}{2} = 2.125$$

Since $f(x_3) = 0.3457$, the root lies between 2 and 2.125.
Therefore,

$$x_4 = \frac{2+2.125}{2} = 2.0625$$

Proceeding in this way, we obtain the successive approximations:

$$x_5 = 2.09375, \quad x_6 = 2.10938, \quad x_7 = 2.10156,$$
$$x_8 = 2.09766, \quad x_9 = 2.09570, \quad x_{10} = 2.09473,$$
$$x_{11} = 2.09424, \ldots$$

We find

$$x_{11} - x_{10} = -0.0005,$$

and

$$\left|\frac{x_{11} - x_{10}}{x_{11}}\right| \times 100 = \frac{0.0005}{2.09424} \times 100 = 0.02\%$$

Hence a root, correct to three decimal places, is 2.094.

Example 2.3 Find a real root of $f(x) = x^3 + x^2 + x + 7 = 0$ correct to three decimal places.

The given equation is a cubic and the last term is positive. Hence, $f(x) = 0$ will have a negative real root. We find that

$$f(-1) = 6, \quad f(-2) = 1 \text{ and } f(-3) = -14.$$

Therefore, a real root lies between –3 and –2.

We take

$$x_1 = \frac{-2-3}{2} = -2.5$$

Since $f(-2.5) = -4.875$, the root lies between –2 and –2.5, and then

$$x_2 = \frac{-2-2.5}{2} = -2.25$$

Now $f(x_2) = -1.5781$, and, therefore, the root lies between –2 and –2.25.

It follows that

$$x_3 = \frac{-4.25}{2} = -2.125$$

Successive approximations are given by

$$x_4 = -2.0625, \quad x_5 = -2.0938, \quad x_6 = -2.1094,$$
$$x_7 = -2.1016, \quad x_8 = -2.1055, \quad x_9 = -2.1035,$$
$$x_{10} = -2.1045, \quad x_{11} = -2.1050, \ldots$$

The difference between x_{10} and x_{11} is 0.0005. Hence, we conclude that the root is given by $x = -2.105$, correct to three decimal places.

Example 2.4 Find the positive root, between 0 and 1, of the equation $x = e^{-x}$ to a tolerance of 0.05%.

Let

$$f(x) = xe^x - 1 = 0$$

We have, $f(0) = -1$ and $f(1) = e - 1$, which is positive. Hence, a root exists between 0 and 1, and

$$x_1 = \frac{0+1}{2} = 0.5$$

Because, $f(x_1) = -0.1756$, the root lies between 0.5 and 1.0.

Then

$$x_2 = \frac{0.5+1.0}{2} = 0.75$$

Now, the tolerance ε_1 is given by

$$\varepsilon_1 = \left|\frac{x_2 - x_1}{x_2}\right| \times 100$$

$$= \frac{0.25}{0.75} \times 100 = 33.33\%$$

since $f(x_2) = 0.5878$, the root lies between 0.5 and 0.75.
Therefore,

$$x_3 = \frac{0.5 + 0.75}{2} = 0.625$$

also,

$$\varepsilon_2 = \left|\frac{0.625 - 0.75}{0.625}\right| \times 100 = 20\%.$$

Proceeding in this way, successive approximations and tolerances are obtained:

$x_4 = 0.5625$, $\varepsilon_3 = 11.11\%$; $x_5 = 0.5938$, $\varepsilon_4 = 5.26\%$;
$x_6 = 0.5781$, $\varepsilon_5 = 2.71\%$; $x_7 = 0.5703$, $\varepsilon_6 = 1.37\%$;
$x_8 = 0.5664$, $\varepsilon_7 = 0.69\%$; $x_9 = 0.5684$, $\varepsilon_8 = 0.35\%$;
$x_{10} = 0.5674$, $\varepsilon_9 = 0.18\%$; $x_{11} = 0.5669$, $\varepsilon_{10} = 0.09\%$;
$x_{12} = 0.5671$, $\varepsilon_{11} = 0.035\%$

Since $\varepsilon_{11} = 0.035\% < 0.05\%$, the required root is 0.567, correct to three decimal places.

Example 2.5 Find a root, correct to three decimal places and lying between 0 and 0.5, of the equation

$$4e^{-x} \sin x - 1 = 0$$

Let

$$f(x) = 4e^{-x} \sin x - 1$$

We have $f(0) = -1$ and $f(0.5) = 0.163145$
Therefore,

$$x_1 = 0.25$$

Since $f(0.25) = -0.22929$, it follows that the root lies between 0.25 and 0.5.
Therefore,

$$x_2 = \frac{0.75}{2} = 0.375$$

The successive approximations are given by

$x_3 = 0.3125$, $x_4 = 0.3438$, $x_5 = 0.3594$,
$x_6 = 0.3672$, $x_7 = 0.3711$, $x_8 = 0.3692$,
$x_9 = 0.3702$, $x_{10} = 0.3706$, $x_{11} = 0.3704$
$x_{12} = 0.3705$, …

Hence the required root is 0.371, correct to three decimal places.

2.3 METHOD OF FALSE POSITION

This is the oldest method for finding the real root of a nonlinear equation $f(x) = 0$ and closely resembles the bisection method. In this method, also known as *regula–falsi* or the *method of chords*, we choose two points a and b such that $f(a)$ and $f(b)$ are of opposite signs. Hence, a root must lie in between these points. Now, the equation of the chord joining the two points $[a, f(a)]$ and $[b, f(b)]$ is given by

$$\frac{y - f(a)}{x - a} = \frac{f(b) - f(a)}{b - a}. \tag{2.6}$$

The method consists in replacing the part of the curve between the points $[a, f(a)]$ and $[b, f(b)]$ by means of the *chord* joining these points, and taking the point of intersection of the chord with the x-axis as an *approximation* to the root. The point of intersection in the present case is obtained by putting $y = 0$ in Eq. (2.6). Thus, we obtain

$$x_1 = a - \frac{f(a)}{f(b) - f(a)}(b - a) = \frac{af(b) - bf(a)}{f(b) - f(a)}, \tag{2.7}$$

which is the *first approximation* to the root of $f(x) = 0$. If now $f(x_1)$ and $f(a)$ are of opposite signs, then the root lies between a and x_1, and we replace b by x_1 in Eq. (2.7), and obtain the *next* approximation. Otherwise, we replace a by x_1 and generate the next approximation. The procedure is repeated till the root is obtained to the desired accuracy. Figure 2.2 gives a graphical representation of the method. The error criterion Eq. (2.5) can be used in this case also.

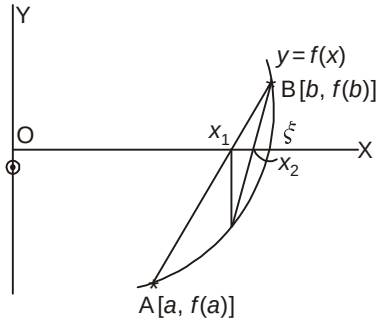


Figure 2.2 Method of false position.

Example 2.6 Find a real root of the equation:

$$f(x) = x^3 - 2x - 5 = 0.$$

We find $f(2) = -1$ and $f(3) = 16$. Hence $a = 2$, $b = 3$, and a root lies between 2 and 3. Equation (2.7) gives

$$x_1 = \frac{2(16) - 3(-1)}{16 - (-1)} = \frac{35}{17} = 2.058823529.$$

Now, $f(x_1) = -0.390799917$ and hence the root lies between 2.058823529 and 3.0. Using formula (2.7), we obtain

$$x_2 = \frac{2.058823529(16) - 3(-0.390799917)}{16.390799917} = 2.08126366.$$

Since $f(x_2) = -0.147204057$, it follows that the root lies between 2.08126366 and 3.0. Hence, we have

$$x_3 = \frac{2.08126366(16) - 3(-0.147204057)}{16.147204057} = 2.089639211.$$

Proceeding in this way, we obtain successively:

$$x_4 = 2.092739575, \qquad x_5 = 2.09388371,$$

$$x_6 = 2.094305452, \qquad x_7 = 2.094460846,\ldots$$

The correct value is 2.0945…, so that x_7 is correct to five significant figures.

Example 2.7 Given that the equation $x^{2.2} = 69$ has a root between 5 and 8. Use the method of regula–falsi to determine it.

Let $f(x) = x^{2.2} - 69$. We find

$$f(5) = -34.50675846 \quad \text{and} \quad f(8) = 28.00586026.$$

Hence

$$x_1 = \frac{5(28.00586026) - 8(-34.50675846)}{28.00586026 + 34.50675846} = 6.655990062.$$

Now, $f(x_1) = -4.275625415$ and therefore, the root lies between 6.655990062 and 8.0. We obtain

$$x_2 = 6.83400179, \qquad x_3 = 6.850669653.$$

The correct root is 6.8523651…, so that x_3 is correct to three significant figures.

Example 2.8 The equation $2x = \log_{10} x + 7$ has a root between 3 and 4. Find this root, correct to three decimal places, by regula–falsi method.

Let

$$f(x) = 2x - \log_{10} x - 7,\ a = 3 \text{ and } b = 4.$$

Then we find

$$f(3) = -1.4771 \text{ and } f(4) = 0.3979.$$

Hence

$$x_1 = \frac{af(b) - bf(a)}{f(b) - f(a)}$$
$$= \frac{3(0.3979) - 4(-1.4771)}{0.3979 + 1.4771}$$
$$= \frac{7.1021}{1.8750} = 3.7878.$$

Therefore, the root lies between 3 and 3.7878. Now, we take $a = 3$ and $b = 3.7878$ Then,

$$f(b) = 2(3.7878) - \log_{10} 3.7878 - 7 = -0.002787$$

Hence,

$$x_2 = \frac{3(-0.002787) - 3.7878(-1.4771)}{-0.002787 + 1.4771}$$
$$= 3.7893,$$

and

$$f(x_2) = 2(3.7893) - \log_{10}(3.7893) - 7$$
$$= 0.000041,$$

which shows that $x = 3.789$ is the root correct to three decimal places.

Example 2.9 Find a root of the equation $4e^{-x} \sin x - 1 = 0$ by regular–falsi method given that the root lies between 0 and 0.5.

Let

$$f(x) = 4e^{-x} \sin x - 1, \; a = 0, \; b = 0.5.$$

We have

$$f(a) = -1 \text{ and } f(b) = 4e^{-0.5} \sin 0.5 - 1 = 0.163145$$

Therefore,

$$x_1 = \frac{0(0.163145) - 0.5(-1)}{1.163145}$$
$$= \frac{0.5}{1.163145} = 0.4298690$$

Now, we take

$$a = 0 \quad \text{and} \quad b = 0.4298690$$

Then

$$f(x) = 0.08454$$

Therefore,

$$x_2 = \frac{0(0.08454) - 0.42987(-1)}{1.08454}$$
$$= 0.39636$$

Now,

$$a = 0, \quad b = 0.39636 \quad \text{and} \quad f(b) = 0.038919$$

Hence

$$x_3 = \frac{0(0.038919) - 0.39636(-1)}{1.038919}$$
$$= 0.381512,$$

and

$$f(x_3) = 0.016934$$

Taking $a = 0$ and $b = 0.381512$, we obtain

$$x_4 = \frac{0(0.016934) - 0.381512(-1)}{1.016934}$$
$$= 0.375159,$$

and

$$f(x_4) = 0.0071873$$

Proceeding as above, we obtain

$$x_5 = 0.37248, \qquad x_6 = 0.37136,$$
$$x_7 = 0.37089, \qquad x_8 = 0.370697$$

It follows that the required root is 0.371, correct to three decimal places.

2.4 ITERATION METHOD

We have so far discussed root-finding methods which require an interval in which the root lies. We now describe methods which require one or more approximate values to start the solution and these values need not necessarily bracket the root. The first is the *iteration* method which requires one starting value of x.

To describe this method for finding a root of the equation

$$f(x) = 0, \tag{2.1}$$

we rewrite this equation in the form

$$x = \phi(x) \tag{2.8}$$

There are many ways of doing this. For example, the equation

$$x^3 + x^2 - 2 = 0$$

can be expressed in different forms

$$x = \sqrt{\frac{2}{1+x}}, \quad x = \sqrt{2 - x^3}, \quad x = (2 - x^2)^{1/3}, \text{ etc.}$$

Now, let x_0 be an approximate root of Eq. (2.8). Then, substituting in Eq. (2.8), we get the first approximation as

$$x_1 = \phi(x_0)$$

Successive substitutions give the approximations

$$x_2 = \phi(x_1), \qquad x_3 = \phi(x_2), \ldots, \qquad x_n = \phi(x_{n-1}).$$

The preceding sequence may not converge to a definite number. But if the sequence converges to a definite number ξ, then ξ will be a root of the equation $x = \phi(x)$. To show this, let

$$x_{n+1} = \phi(x_n) \tag{2.9}$$

be the relation between the nth and $(n + 1)$th approximations. As n increases, $x_{n+1} \to \xi$ and if $\phi(x)$ is a continuous function, then $\phi(x_n) \to \phi(\xi)$. Hence, in the limit, we obtain

$$\xi = \phi(\xi), \tag{2.10}$$

which shows that ξ is a root of the equation $x = \phi(x)$.

To establish the condition of convergence of Eq. (2.8), we proceed in the following way:

From Eq. (2.9), we have

$$x_1 = \phi(x_0) \tag{2.11}$$

From Eqs. (2.10) and (2.11), we get

$$\begin{aligned} \xi - x_1 &= \phi(\xi) - \phi(x_0) \\ &= (\xi - x_0)\,\phi'(\xi_0), \quad x_0 < \xi_0 < \xi, \end{aligned} \tag{2.12}$$

on using Theorem 1.5. Similarly, we obtain

$$\xi - x_2 = (\xi - x_1)\ \phi'(\xi_1), \quad x_1 < \xi_1 < \xi \tag{2.13}$$

$$\xi - x_3 = (\xi - x_2)\ \phi'(\xi_2), \quad x_2 < \xi_2 < \xi \tag{2.14}$$

$$\vdots \qquad \vdots \qquad \vdots$$

$$\xi - x_{n+1} = (\xi - x_n)\ \phi'(\xi_n), \quad x_n < \xi_n < \xi \tag{2.15}$$

If we assume

$$|\phi'(\xi_i)| \le k \text{ (for all } i), \tag{2.16}$$

then Eqs. (2.12) to (2.15) give

$$\left.\begin{aligned} |\xi - x_1| &\le k\,|\xi - x_0| \\ |\xi - x_2| &\le k\,|\xi - x_1| \\ |\xi - x_3| &\le k\,|\xi - x_2| \\ &\vdots \qquad\qquad \vdots \\ |\xi - x_{n+1}| &\le k\,|\xi - x_n| \end{aligned}\right\} \tag{2.17}$$

Multiplying the corresponding sides of the above equations, we obtain

$$|\xi - x_{n+1}| \le k^{n+1}\,|\xi - x_0| \tag{2.18}$$

If $k < 1$, i.e., if $|\phi'(\xi_i)| < 1$, then the right side of Eq. (2.18) tends to zero and the sequence of approximations $x_0, x_1, x_2, \ldots$ converges to the root ξ. Thus, when we express the equation $f(x) = 0$ in the form $x = \phi(x)$, then $\phi(x)$ must be such that

$$|\,\phi'(x)\,| < 1$$

in an immediate neighbourhood of the root. It follows that if *the initial approximation x_0 is chosen in an interval containing the root ξ, then the sequence of approximations converges to the root ξ.*

Now, we show that the root so obtained is *unique*. To prove this, let ξ_1 and ξ_2 be two roots of the equation $x = \phi(x)$. Then, we must have

$$\xi_1 = \phi(\xi_1) \quad \text{and} \quad \xi_2 = \phi(\xi_2).$$

Therefore,

$$\begin{aligned} |\xi_1 - \xi_2| &= |\phi(\xi_1) - \phi(\xi_2)| \\ &= |\xi_1 - \xi_2|\,\phi'(\eta),\ \eta \in (\xi_1, \xi_2) \end{aligned}$$

Hence,

$$|\xi_1 - \xi_2|\,[1 - \phi'(\eta)] = 0 \tag{2.19}$$

Since $|\phi'(\eta)| < 1$, it follows that $\xi_1 = \xi_2$, which proves that the root obtained is unique.

Finally, we shall find the error in the root obtained. We have

$$\begin{aligned} |\xi - x_n| &\le k|\xi - x_{n-1}| \\ &= k|\xi - x_n + x_n - x_{n-1}| \\ &\le k\left[|\xi - x_n| + |x_n - x_{n-1}|\right] \end{aligned}$$

$$\Rightarrow \qquad |\xi - x_n| \le \frac{k}{1-k}|x_n - x_{n-1}| = \frac{k}{1-k}k^{n-1}|x_1 - x_0|$$

$$\le \frac{k^n}{1-k}|x_1 - x_0|, \tag{2.20}$$

which shows that the convergence would be faster for smaller values of k.

Now, let ε be the specified accuracy so that

$$|\xi - x_n| \le \varepsilon$$

Then, Eq. (2.20) gives

$$|x_n - x_{n-1}| \le \frac{1-k}{k}\varepsilon, \tag{2.21}$$

which can be used to find the difference between two successive approximations (or *iterations*) to achieve a prescribed accuracy.

Example 2.10 Find a real root of the equation $x^3 = 1 - x^2$ on the interval [0, 1] with an accuracy of 10^{-4}.

We rewrite the equation as

$$x = \frac{1}{\sqrt{x+1}} \tag{i}$$

Here

$$\phi(x) = \frac{1}{\sqrt{x+1}} = (x+1)^{-1/2}$$

Therefore,

$$\phi'(x) = -\frac{1}{2}(x+1)^{-3/2} = -\frac{1}{2\sqrt{(x+1)^3}} < 1 \text{ in } [0, 1].$$

Also,

$$\max|\phi'(x)| = \frac{1}{2\sqrt{8}} = \frac{1}{4\sqrt{2}} = k < 0.2$$

Therefore, Eq. (2.21) gives

$$|x_n - x_{n-1}| \le \frac{1-0.2}{0.2}\varepsilon = 4\times10^{-4} = 0.0004.$$

Taking $x_0 = 0.75$, we find

$$x_1 = \frac{1}{\sqrt{1.75}} = 0.75593,$$

$$x_2 = \frac{1}{\sqrt{1.75593}} = 0.75465,$$

$$x_3 = \frac{1}{\sqrt{1.75465}} = 0.75493.$$

Now, $|x_3 - x_2| = 0.00028 < 0.0004$. Hence, the required root is 0.7549, correct to four decimal places.

Example 2.11 Find a real root, correct to three decimal places, of the equation

$$2x - 3 = \cos x$$

lying in the interval $\left[\frac{3}{2}, \frac{\pi}{2}\right]$.

We rewrite the given equation as

$$x = \frac{1}{2}(\cos x + 3)$$

Here

$$|\phi'(x)| = \frac{1}{2}|\sin x| < 1, \text{ in } \left[\frac{3}{2}, \frac{\pi}{2}\right].$$

Choosing $x_0 = 1.5$, we obtain successively:

$$x_1 = \frac{1}{2}(\cos 1.5 + 3) = 1.5354,$$

$$x_2 = \frac{1}{2}(\cos 1.5354 + 3) = 1.5177,$$

$$x_3 = \frac{1}{2}(\cos 1.5177 + 3) = 1.5265,$$

$$x_4 = \frac{1}{2}(\cos 1.5265 + 3) = 1.5221,$$

$$x_5 = \frac{1}{2}(\cos 1.5221 + 3) = 1.5243,$$

$$x_6 = \frac{1}{2}(\cos 1.5243 + 3) = 1.5232,$$

$$x_7 = \frac{1}{2}(\cos 1.5232 + 3) = 1.5238.$$

Now, $| x_7 - x_6 | = 0.0006 < 0.001$. Hence, the root, correct to three decimal places is 1.524.

Example 2.12 Use the method of iteration to find a positive root of the equation $xe^x = 1$, given that a root lies between 0 and 1.

Writing the equation in the form

$$x = e^{-x},$$

we find that

$$\phi'(x) = -e^{-x} = -\frac{1}{e} \qquad \text{for } x = 1$$

Therefore,

$$| \phi'(x) | < 1.$$

Choosing $x_0 = 0.5$, we find

$x_1 = e^{-0.5} = 0.60653,$ $x_2 = 0.54524,$
$x_3 = 0.57970,$ $x_4 = 0.56007,$
$x_5 = 0.57117,$ $x_6 = 0.56486,$
$x_7 = 0.56844,$ $x_8 = 0.56641,$
$x_9 = 0.56756,$ $x_{10} = 0.56691,$
$x_{11} = 0.56728,$ $x_{12} = 0.56706,$
$x_{13} = 0.56719,$ $x_{14} = 0.56712,$
$x_{15} = 0.56716,$ $x_{16} = 0.56713,$
$x_{17} = 0.56715,$ $x_{18} = 0.56714,$
$x_{19} = 0.56714.$

It follows that the root, correct to four decimal places, is 0.5671.

Example 2.13 Use the iterative method to find a real root of the equation $\sin x = 10(x - 1)$. Give your answer correct to three decimal places.

Let

$$f(x) = \sin x - 10x + 10$$

We find, from graph, that a root lies between 1 and π (The student is advised to draw the graphs). Rewriting the equation as

$$x = 1 + \frac{\sin x}{10},$$

we have

$$\phi(x) = 1 + \frac{\sin x}{10},$$

and

$$|\phi'(x)| = \frac{\cos x}{10} < 1 \text{ in } 1 \le x \le \pi.$$

Taking $x_0 = 1$, we obtain the successive iterates as:

$$x_1 = 1 + \frac{\sin 1}{10} = 1.0841,$$

$$x_2 = 1 + \frac{\sin 1.0841}{10} = 1.0884,$$

$$x_3 = 1 + \frac{\sin 1.0884}{10} = 1.0886,$$

$$x_4 = 1 + \frac{\sin 1.0886}{10} = 1.0886.$$

Hence the required root is 1.089.

Acceleration of convergence: Aitken's Δ^2-process

From the relation

$$|\xi - x_{n+1}| = |\phi(\xi) - \phi(x_n)| \le k\,|\xi - x_n|, \qquad k < 1$$

it is clear that the iteration method is linearly convergent. This slow rate of convergence can be accelerated by using Aitken's method, which is described below.

Let x_{i-1}, x_i, x_{i+1} be three successive approximations to the desired root $x = \xi$ of the equation $x = \phi(x)$. From Eq. (2.17), we know that

$$\xi - x_i = k(\xi - x_{i-1}), \qquad \xi - x_{i+1} = k(\xi - x_i)$$

Dividing, we obtain

$$\frac{\xi - x_i}{\xi - x_{i+1}} = \frac{\xi - x_{i-1}}{\xi - x_i},$$

which gives on simplification

$$\xi = x_{i+1} - \frac{(x_{i+1} - x_i)^2}{x_{i+1} - 2x_i + x_{i-1}}. \tag{2.22}$$

If we now define Δx_i and $\Delta^2 x_i$ by the relations

$$\Delta x_i = x_{i+1} - x_i \quad \text{and} \quad \Delta^2 x_i = \Delta(\Delta x_i),$$

then

$$\begin{aligned}
\Delta^2 x_{i-1} &= \Delta(\Delta x_{i-1}) \\
&= \Delta(x_i - x_{i-1}) \\
&= \Delta x_i - \Delta x_{i-1} \\
&= x_{i+1} - x_i - (x_i - x_{i-1}) \\
&= x_{i+1} - 2x_i + x_{i-1}.
\end{aligned}$$

Hence Eq. (2.22) can be written in the simpler form

$$\xi = x_{i+1} - \frac{(\Delta x_i)^2}{\Delta^2 x_{i-1}}. \tag{2.23}$$

which explains the term Δ^2-process.

In any numerical application, the values of the following underlined quantities must be obtained.

x_{i-1}		
	Δx_{i-1}	
x_i		$\underline{\Delta^2 x_{i-1}}$
	$\underline{\Delta x_i}$	
$\underline{x_{i+1}}$		

Example 2.14 We consider again Example 2.11, viz., the equation

$$x = \frac{1}{2}(3 + \cos x)$$

As before,

$x_1 = 1.5$		
	0.035	
$x_2 = 1.535$		-0.052
	-0.017	
$x_3 = 1.518$		

Hence we obtain from Eq. (2.23)

$$x_4 = 1.518 - \frac{(-0.017)^2}{-0.052} = 1.524,$$

which corresponds to six normal iterations.

2.5 NEWTON–RAPHSON METHOD

This method is generally used to improve the result obtained by one of the previous methods. Let x_0 be an approximate root of $f(x) = 0$ and let $x_1 = x_0 + h$ be the correct root so that $f(x_1) = 0$. Expanding $f(x_0 + h)$ by Taylor's series, we obtain

$$f(x_0) + hf'(x_0) + \frac{h^2}{2!} f''(x_0) + \cdots = 0.$$

Neglecting the second and higher-order derivatives, we have

$$f(x_0) + hf'(x_0) = 0,$$

which gives

$$h = -\frac{f(x_0)}{f'(x_0)}.$$

A better approximation than x_0 is, therefore, given by x_1, where

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}. \tag{2.24a}$$

Successive approximations are given by $x_2, x_3, \ldots, x_{n+1}$, where

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, \tag{2.24b}$$

which is the *Newton–Raphson formula*.

If we compare Eq. (2.24b) with the relation

$$x_{n+1} = \phi(x_n)$$

of the iterative method [see Eq. (2.9)], we obtain

$$\phi(x) = x - \frac{f(x)}{f'(x)},$$

which gives

$$\phi'(x) = \frac{f(x)\, f''(x)}{[f'(x)]^2}. \tag{2.25}$$

To examine the convergence we assume that $f(x)$, $f'(x)$ and $f''(x)$ are continuous and bounded on any interval containing the root $x = \xi$ of the equation $f(x) = 0$. If ξ is a simple root, then $f'(x) \neq 0$. Further since $f'(x)$ is continuous, $|f'(x)| \geq \varepsilon$ for some $\varepsilon > 0$ in a suitable neighbourhood of ξ. Within this neighbourhood we can select an interval such that $|f(x)\, f''(x)| < \varepsilon^2$ and this is possible since $f(\xi) = 0$ and since $f(x)$ is continuously twice differentiable. Hence, in this interval we have

$$|\phi'(x)| < 1. \tag{2.26}$$

Therefore, Newton–Raphson formula given in Eq. (2.24b) converges, provided that the initial approximation x_0 is chosen sufficiently close to ξ. When ξ is a multiple root, the Newton–Raphson method still converges but slowly. Convergence can, however, be made faster by modifying Eq. (2.24b). This will be discussed later.

To obtain the rate of convergence of the method, we note that $f(\xi) = 0$ so that Taylor's expansion gives

$$f(x_n) + (\xi - x_n) f'(x_n) + \frac{1}{2}(\xi - x_n)^2 f''(x_n) + \cdots = 0,$$

from which we obtain

$$-\frac{f(x_n)}{f'(x_n)} = (\xi - x_n) + \frac{1}{2}(\xi - x_n)^2 \frac{f''(x_n)}{f'(x_n)} \tag{2.27}$$

From Eqs. (2.24b) and (2.27), we have

$$x_{n+1} - \xi = \frac{1}{2}(x_n - \xi)^2 \frac{f''(x_n)}{f'(x_n)} \tag{2.28}$$

Setting

$$\varepsilon_n = x_n - \xi, \tag{2.29}$$

Equation (2.28) gives

$$\varepsilon_{n+1} \approx \frac{1}{2}\varepsilon_n^2 \frac{f''(\xi)}{f'(\xi)}, \tag{2.30}$$

so that the Newton–Raphson process has a second-order or quadratic convergence.

Geometrically, the method consists in replacing the part of the curve between the point $[x_0, f(x_0)]$ and the x-axis by means of the tangent to the curve at the point, and is described graphically in Fig. 2.3. It can be used for solving both algebraic and transcendental equations and it can also be used when the roots are complex.

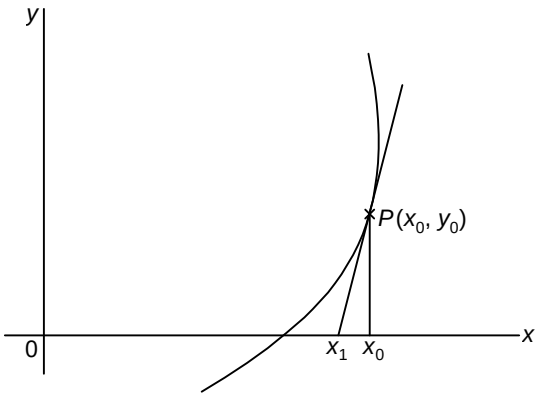


Figure 2.3 Newton–Raphson method.

Example 2.15 Use the Newton–Raphson method to find a root of the equation $x^3 - 2x - 5 = 0$.

Here $f(x) = x^3 - 2x - 5$ and $f'(x) = 3x^2 - 2$. Hence Eq. (2.24b) gives:

$$x_{n+1} = x_n - \frac{x_n^3 - 2x_n - 5}{3x_n^2 - 2} \tag{i}$$

Choosing $x_0 = 2$, we obtain $f(x_0) = -1$ and $f'(x_0) = 10$. Putting $n = 0$ in Eq. (i), we obtain

$$x_1 = 2 - \left(-\frac{1}{10}\right) = 2.1$$

Now,

$$f(x_1) = (2.1)^3 - 2(2.1) - 5 = 0.061,$$

and

$$f'(x_1) = 3(2.1)^2 - 2 = 11.23.$$

Hence

$$x_2 = 2.1 - \frac{0.061}{11.23} = 2.094568.$$

This example demonstrates that Newton–Raphson method converges more rapidly than the methods described in the previous sections, since this requires fewer iterations to obtain a specified accuracy. But since two function evaluations are required for each iteration, Newton–Raphson method requires more computing time.

Example 2.16 Find a root of the equation $x \sin x + \cos x = 0$.

We have

$$f(x) = x \sin x + \cos x \quad \text{and} \quad f'(x) = x \cos x.$$

The iteration formula is, therefore,

$$x_{n+1} = x_n - \frac{x_n \sin x_n + \cos x_n}{x_n \cos x_n}.$$

With $x_0 = \pi$, the successive iterates are given below

n	x_n	$f(x_n)$	x_{n+1}
0	3.1416	–1.0	2.8233
1	2.8233	–0.0662	2.7986
2	2.7986	–0.0006	2.7984
3	2.7984	0.0	2.7984

Example 2.17 Find a real root of the equation $x = e^{-x}$, using the Newton–Raphson method.

We write the equation in the form

$$f(x) = xe^x - 1 = 0 \tag{i}$$

Let $x_0 = 1$. Then

$$x_1 = 1 - \frac{e-1}{2e} = \frac{1}{2}\left(1 + \frac{1}{e}\right) = 0.6839397$$

Now

$$f(x_1) = 0.3553424, \quad \text{and} \quad f'(x_1) = 3.337012,$$

so that

$$x_2 = 0.6839397 - \frac{0.3553424}{3.337012} = 0.5774545$$

Proceeding in this way, we obtain

$$x_3 = 0.5672297 \quad \text{and} \quad x_4 = 0.5671433.$$

Example 2.18 Using Newton–Raphson method, find a real root, correct to 3 decimal places, of the equation $\sin x = x/2$ given that the root lies between $\pi/2$ and π.

Let

$$f(x) = \sin x - \frac{x}{2}$$

Then

$$f'(x) = \cos x - \frac{1}{2}$$

Choosing $x_0 = \frac{\pi}{2}$, we obtain

$$x_1 = \frac{\pi}{2} - \frac{\sin\frac{\pi}{2} - \frac{\pi}{4}}{-\frac{1}{2}} = 2,$$

$$x_2 = 2 - \frac{\sin 2 - 1}{\cos 2 - \frac{1}{2}} = 1.9010,$$

$$x_3 = 1.9010 - \frac{\sin 1.9010 - 0.9505}{\cos 1.9010 - 0.5} = 1.8955$$

Similarly, $x_4 = 1.8954$, $x_5 = 1.8955$, … . Hence the required root is $x = 1.896$.

Example 2.19 Given the equation $4e^{-x} \sin x - 1 = 0$, find the root between 0 and 0.5 correct to three decimal places.

Let $f(x) = 4e^{-x} \sin x - 1$ and $x_0 = 0.2$.

Then

$$f(x_0) = -0.349373236,$$

and

$$f'(x_0) = 2.559015826.$$

Therefore,

$$x_1 = 0.2 + \frac{0.349373236}{2.559015826}$$
$$= 0.336526406 = 0.33653.$$

Now,

$$f(x_1) = -0.056587$$

and

$$f'(x_1) = 1.753305735$$
$$= 1.75330$$

Therefore,

$$x_2 = 0.33653 + \frac{0.056587}{1.75330} = 0.36880$$

For the next approximation, we find $f(x_2) = -0.00277514755$ and $f'(x_2) = 1.583028705$. This gives

$$x_3 = 0.36880 - 0.00175 = 0.37055$$

since $f(x_3) = -0.00001274$, it follows that the required root is given by $x = 0.370$.

Generalized Newton's method

If ξ is a root of $f(x) = 0$ with multiplicity p, then the iteration formula corresponding to Eq. (2.24) is taken as

$$x_{n+1} = x_n - p\frac{f(x_n)}{f'(x_n)}, \tag{2.31}$$

which means that $(1/p)f'(x_n)$ is the slope of the straight line passing through (x_n, y_n) and intersecting the x-axis at the point $(x_{n+1}, 0)$.

Equation (2.31) is called the *generalized Newton's formula* and reduces to Eq. (2.24) for $p = 1$. Since ξ is a root of $f(x) = 0$ with multiplicity p, it follows that ξ is also a root of $f'(x) = 0$ with multiplicity $(p - 1)$, of $f''(x) = 0$ with multiplicity $(p - 2)$, and so on. Hence the expressions

$$x_0 - p\frac{f(x_0)}{f'(x_0)}, \quad x_0 - (p-1)\frac{f'(x_0)}{f''(x_0)}, \quad x_0 - (p-2)\frac{f''(x_0)}{f'''(x_0)}$$

must have the same value if there is a root with multiplicity p, provided that the initial approximation x_0 is chosen sufficiently close to the root.

Example 2.20 Find a double root of the equation $f(x) = x^3 - x^2 - x + 1 = 0$.

Choosing $x_0 = 0.8$, we have

$$f'(x) = 3x^3 - 2x - 1, \quad \text{and} \quad f''(x) = 6x - 2.$$

With $x_0 = 0.8$, we obtain

$$x_0 - 2\frac{f(x_0)}{f'(x_0)} = 0.8 - 2\frac{0.072}{-(0.68)} = 1.012,$$

and

$$x_0 - \frac{f'(x_0)}{f''(x_0)} = 0.8 - \frac{(-0.68)}{2.8} = 1.043.$$

The closeness of these values indicates that there is a double root near to unity. For the next approximation, we choose $x_1 = 1.01$ and obtain

$$x_1 - 2\frac{f(x_1)}{f'(x_1)} = 1.01 - 0.0099 = 1.0001,$$

and

$$x_1 - \frac{f'(x_1)}{f''(x_1)} = 1.01 - 0.0099 = 1.0001.$$

We conclude, therefore, that there is a double root at $x = 1.0001$ which is sufficiently close to the actual root unity.

On the other hand, if we apply Newton–Raphson method with $x_0 = 0.8$, we obtain

$$x_1 = 0.8 + 0.106 \approx 0.91, \quad \text{and} \quad x_2 = 0.91 + 0.046 \approx 0.96.$$

It is clear that the generalized Newton's method converges more rapidly than the Newton–Raphson procedure.

2.6 RAMANUJAN'S METHOD

Srinivasa Ramanujan (1887–1920) described an iterative procedure* to determine the *smallest* root of the equation

$$f(x) = 0, \tag{2.1}$$

where $f(x)$ is of the form

$$f(x) = 1 - (a_1x + a_2x^2 + a_3x^3 + \cdots) \tag{2.32}$$

To explain the method of procedure, we consider the quadratic equation

$$f(x) = a_0x^2 + a_1x + a_2 = 0,$$

with the roots x_1 and x_2, such that $|x_1| < |x_2|$. Then the equation defined by

$$\phi(x) = a_2x^2 + a_1x + a_0 = 0$$

$$\Rightarrow \qquad 1 + \frac{a_1}{a_0}x + \frac{a_2}{a_0}x^2 = 0$$

will have roots $\frac{1}{x_1}$ and $\frac{1}{x_2}$ such that $\frac{1}{|x_1|} > \frac{1}{|x_2|}$.

Now,

$$\frac{1}{\phi(x)} = \left(1 + \frac{a_1}{a_0}x + \frac{a_2}{a_0}x^2\right)^{-1}$$

*See Berndt [1985], p. 41.

can be written as

$$\left(1+\frac{a_1}{a_0}x+\frac{a_2}{a_0}x^2\right)^{-1} = \frac{k_1}{x-\frac{1}{x_1}}+\frac{k_2}{x-\frac{1}{x_2}}$$

$$= \frac{-k_1x_1}{1-xx_1}+\frac{-k_2x_2}{1-xx_2}$$

$$= -k_1x_1(1-xx_1)^{-1}-k_2x_2(1-xx_2)^{-1}$$

$$= \sum_{i=0}^{\infty} b_i x^i, \quad \text{where } b_i = -\sum_{r=1}^{2} k_r x_r^{i+1}$$

Then,

$$\frac{b_{i-1}}{b_i} = \frac{k_1x_1^i+k_2x_2^i}{k_1x_1^{i+1}+k_2x_2^{i+1}} = \frac{\frac{k_1}{k_2}\left(\frac{x_1}{x_2}\right)^i+1}{\frac{k_1}{k_2}\left(\frac{x_1}{x_2}\right)^{i+1}+1}\cdot\frac{1}{x_2}$$

Since $\frac{x_1}{x_2}<1$, it follows that

$$\lim_{i\to\infty}\frac{b_{i-1}}{b_i}=\frac{1}{x_2},$$

which is the smallest root. This is the basis of Ramanujan's method which is outlined below.

To find the smallest root of $f(x) = 0$, we consider $f(x)$ in the form

$$f(x) = 1 - (a_1x + a_2x^2 + a_3x^3 + \cdots),$$

and then write

$$\left[1-(a_1x+a_2x^2+a_3x^3+\cdots)\right]^{-1} = b_1+b_2x+b_3x^2+\cdots \tag{2.33}$$

$$\Rightarrow \quad 1 + (a_1x + a_2x^2 + a_3x^3 + \cdots) + (a_1x + a_2x^2 + a_3x^3 + \cdots)^2 + \cdots$$
$$= b_1 + b_2x + b_3x^2 + \cdots \tag{2.34}$$

To find b_i, we equate coeffecients of like powers of x on both sides of Eq. (2.34). We then obtain

$$\left.\begin{aligned} b_1 &= 1 \\ b_2 &= a_1 = a_1b_1, \quad \text{since } b_1 = 1 \\ b_3 &= a_2 + a_1^2 = a_2b_1 + a_1b_2, \quad \text{since } b_2 = a_1 \\ &\vdots \quad\quad \vdots \quad\quad \vdots \quad\quad \vdots \\ b_k &= a_1b_{k-1} + a_2b_k + \cdots + a_{k-1}b_1 \\ &= a_{k-1}b_1 + a_{k-2}b_2 + \cdots + a_1b_{k-1} \end{aligned}\right\} \tag{2.35}$$

The ratios $\frac{b_{i-1}}{b_i}$, called the *convergents*, approach, in the limit, the *smallest* root of $f(x) = 0$. The method is demonstrated in the following examples.

Example 2.21 Find the smallest root of the equation

$$f(x) = x^3 - 9x^2 + 26x - 24 = 0$$

We have

$$f(x) = 1 - \frac{26}{24}x + \frac{9}{24}x^2 - \frac{1}{24}x^3$$

$$= 1 - \left(\frac{13}{12}x - \frac{3}{8}x^2 + \frac{1}{24}x^3\right)$$

Here

$$a_1 = \frac{13}{12},\ a_2 = -\frac{3}{8},\ a_3 = \frac{1}{24},\ a_4 = a_5 = \cdots = 0$$

Now,

$$b_1 = 1$$

$$b_2 = a_1 = \frac{13}{12} = 1.0833,$$

Therefore,

$$\frac{b_1}{b_2} = \frac{12}{13} = 0.923$$

$$b_3 = a_1 b_2 + a_2 b_1$$

$$= \frac{13}{12}(1.0833) - \frac{3}{8}(1) = 0.7986$$

Therefore,

$$\frac{b_2}{b_3} = \frac{1.0833}{0.7986} = 1.356$$

$$b_4 = a_1 b_3 + a_2 b_2 + a_3 b_1$$

$$= 1.0833\ (0.7986) + \left(-\frac{3}{8}\right)(1.0833) + \frac{1}{24}(1)$$

$$= 0.5007$$

Therefore,

$$\frac{b_3}{b_4} = 1.595$$

$$b_5 = a_1 b_4 + a_2 b_3 + a_3 b_2$$

$$= 1.0833(0.5007) + \left(-\frac{3}{8}\right)(0.7986) + \frac{1}{24}(1.0833)$$

$$= 0.2880$$

Therefore,

$$\frac{b_4}{b_5} = \frac{0.5007}{0.2880} = 1.7382.$$

$$b_6 = a_1 b_5 + a_2 b_4 + a_3 b_3$$
$$= 0.1575$$

Therefore,

$$\frac{b_5}{b_6} = 1.8286.$$

$$b_7 = 0.0835$$

Therefore,

$$\frac{b_6}{b_7} = 1.8862.$$

$$b_8 = 0.0434$$

Therefore,

$$\frac{b_7}{b_8} = 1.9240.$$

$$b_9 = 0.0223$$

Therefore,

$$\frac{b_8}{b_9} = 1.9462.$$

The roots of the given equation are 2, 3 and 4 and it can be seen that the successive convergents approach the value 2.

Example 2.22 Find a root of the equation $xe^x = 1$.

Let

$$xe^x = 1 \tag{i}$$

Expanding e^x in ascending powers of x and simplifying, we can rewrite Eq. (i) as

$$1 = x + x^2 + \frac{x^3}{2} + \frac{x^4}{6} + \frac{x^5}{24} + \cdots \tag{ii}$$

which is of the form of the right side of Eq. (2.32).

Here,

$$a_1 = 1, \quad a_2 = 1, \quad a_3 = \frac{1}{2}, \quad a_4 = \frac{1}{6}, \quad a_5 = \frac{1}{24}, \ \ldots$$

We then have

$$b_1 = 1,$$
$$b_2 = a_2 = 1,$$
$$b_3 = a_1 b_2 + a_2 b_1 = 1 + 1 = 2$$
$$b_4 = a_1 b_3 + a_2 b_2 + a_3 b_1 = 2 + 1 + \frac{1}{2} = \frac{7}{2},$$

$$b_5 = a_1b_4 + a_2b_3 + a_3b_2 + a_4b_1$$

$$= \frac{7}{2} + 2 + \frac{1}{2} + \frac{1}{6}$$

$$= \frac{37}{6} = 6.1667,$$

$$b_6 = \frac{261}{24} = 10.8750;$$

Hence, we obtain

$$b_1/b_2 = 1,$$
$$b_2/b_3 = 0.5,$$
$$b_3/b_4 = 0.5714,$$
$$b_4/b_5 = 0.5676,$$
$$b_5/b_6 = 0.5670.$$

It can be seen that Newton's method (see Example 2.17) gives the value 0.5671433 to this root.

Example 2.23 Find the smallest root, correct to 4 decimal places, of the equation

$$f(x) = 3x - \cos x - 1 = 0.$$

We have

$$f(x) = 1 - 3x + \cos x$$

$$= 1 - 3x + 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

$$= 2 - 3x - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

$$= 2\left[1 - \frac{3}{2}x - \frac{x^2}{4} + \frac{x^4}{48} - \frac{x^6}{1440} + \frac{x^8}{80640} - \cdots\right]$$

Now, let

$$1 - \left(\frac{3}{2}x + \frac{x^2}{4} - \frac{x^4}{48} + \frac{x^6}{1440} - \frac{x^8}{80640}\right)^{-1} = b_1 + b_2x + b_3x^2 + \cdots$$

Here

$$a_1 = \frac{3}{2},\ a_2 = \frac{1}{4},\ a_3 = 0,\ a_4 = -\frac{1}{48},$$

$$a_5 = 0,\ a_6 = \frac{1}{1440},\ a_7 = 0,\ a_8 = -\frac{1}{80640}, \ldots$$

we then obtain,

$b_1 = 1,$ $b_2 = 1.5,$
$b_3 = 2.5,$ $b_4 = 4.125,$
$b_5 = 6.79167,$ $b_6 = 11.18750,$
$b_7 = 18.42778,$ $b_8 = 30.35365$

The successive convergents are

$$\frac{b_1}{b_2} = 0.66667; \qquad \frac{b_2}{b_3} = 0.60000,$$

$$\frac{b_3}{b_4} = 0.60606; \qquad \frac{b_4}{b_5} = 0.60736,$$

$$\frac{b_5}{b_6} = 0.60708; \qquad \frac{b_6}{b_7} = 0.60710,$$

$$\frac{b_7}{b_8} = 0.607102$$

Hence the required root, correct to four decimal places, is 0.6071.

Example 2.24 Using Ramanujan's method, find a real root of the equation

$$1 - x + \frac{x^2}{(2!)^2} - \frac{x^3}{(3!)^2} + \frac{x^4}{(4!)^2} - \cdots = 0$$

Let

$$1 - \left[x - \frac{x^2}{(2!)^2} + \frac{x^3}{(3!)^2} - \frac{x^4}{(4!)^2} + \cdots \right] = 0 \qquad \text{(i)}$$

we have

$$a_1 = 1, \qquad a_2 = -\frac{1}{(2!)^2}, \qquad a_3 = \frac{1}{(3!)^2},$$

$$a_4 = -\frac{1}{(4!)^2}, \qquad a_5 = \frac{1}{(5!)^2}, \qquad a_6 = -\frac{1}{(6!)^2}, \ldots$$

Then we obtain

$$b_1 = 1; \qquad b_2 = a_1 = 1;$$

$$b_3 = a_1 b_2 + a_2 b_1 = 1 - \frac{1}{(2!)^2} = \frac{3}{4};$$

$$b_4 = a_1 b_3 + a_2 b_2 + a_3 b_1$$

$$= \frac{3}{4} - \frac{1}{4} + \frac{1}{36} = \frac{19}{36};$$

$$b_5 = \frac{211}{576}, \ldots$$

The successive convergents are:

$$\frac{b_1}{b_2} = 1; \qquad \frac{b_2}{b_3} = \frac{4}{3} = 1.3333\ldots$$

$$\frac{b_3}{b_4} = \frac{27}{19} = 1.4210\ldots$$

$$\frac{b_4}{b_5} = 1.4408,$$

where the last result is correct to three significant figures.

2.7 SECANT METHOD

We have seen that the Newton–Raphson method requires the evaluation of derivatives of the function and this is not always possible, particularly in the case of functions arising in practical problems. In the secant method, the derivative at x_i is approximated by the formula

$$f'(x_i) \approx \frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}},$$

which can be written as

$$f_i' = \frac{f_i - f_{i-1}}{x_i - x_{i-1}}, \tag{2.36}$$

where $f_i = f(x_i)$. Hence, the Newton–Raphson formula becomes

$$x_{i+1} = x_i - \frac{f_i(x_i - x_{i-1})}{f_i - f_{i-1}} = \frac{x_{i-1} f_i - x_i f_{i-1}}{f_i - f_{i-1}}. \tag{2.37}$$

It should be noted that this formula requires two initial approximations to the root.

Example 2.25 Find a real root of the equation $x^3 - 2x - 5 = 0$ using secant method.

Let the two initial approximations be given by

$$x_{-1} = 2 \quad \text{and} \quad x_0 = 3$$

We have

$$f(x_{-1}) = f_1 = 8 - 9 = -1, \quad \text{and} \quad f(x_0) = f_0 = 27 - 11 = 16.$$

Putting $i = 0$ in Eq. (2.37), we obtain

$$x_1 = \frac{2(16) - 3(-1)}{17} = \frac{35}{17} = 2.058823529.$$

Also,

$$f(x_1) = f_1 = -0.390799923.$$

Putting $i = 1$ in Eq. (2.37), we obtain

$$x_2 = \frac{x_0 f_1 - x_1 f_0}{f_1 - f_0} = \frac{3(-0.390799923) - 2.058823529(16)}{-16.390799923} = 2.08126366.$$

Again

$$f(x_2) = f_2 = -0.147204057.$$

Setting $i = 2$ in Eq. (2.37), and simplifying, we get $x_3 = 2.094824145$, which is correct to three significant figures.

Example 2.26 Using the secant method, find a real root of the equation

$$f(x) = xe^x - 1 = 0$$

We have

$$f(0) = -1 \quad \text{and} \quad f(1) = e - 1 = 1.71828 = f_1$$

Therefore, a root lies between 0 and 1.

Let

$$x_0 = 0 \quad \text{and} \quad x_1 = 1.$$

Therefore,

$$x_2 = \frac{x_0 f_1 - x_1 f_0}{f_1 - f_0} = \frac{1}{2.71828} = 0.36788.$$

and

$$f_2 = 0.36788e^{0.36788} - 1$$
$$= -0.46854.$$

Hence

$$x_3 = \frac{x_1 f_2 - x_2 f_1}{f_2 - f_1}$$
$$= \frac{1(-0.46854) - 0.36788(1.71828)}{-0.46854 - 1.71828}$$
$$= 0.50332$$

and

$$f_3 = -0.16740$$

Hence

$$x_4 = \frac{x_2 f_3 - x_3 f_2}{f_3 - f_2} = 0.57861$$

and

$$f_4 = 0.03198$$

Hence

$$x_5 = \frac{x_3 f_4 - x_4 f_3}{f_4 - f_3} = 0.56653.$$

and

$$f_5 = -0.00169$$

Therefore,

$$x_6 = \frac{x_4 f_5 - x_5 f_4}{f_5 - f_4} = 0.56714.$$

We also find

$$f(x_6) = -0.0001196.$$

It follows that the required root is 0.5671, correct to four decimal places.

2.8 MULLER'S METHOD

In this method, the given function $f(x)$ is approximated by a second degree curve in the vicinity of a root. The roots of the quadratic are then assumed to be the approximations to the roots of the equation $f(x) = 0$. The method is iterative and can be used to compute complex roots. It has quadratic convergence [see, Muller (1956)].

Let (x_{i-2}, y_{i-2}), (x_{i-1}, y_{i-1}) and (x_i, y_i) be three distinct points on the curve $y = f(x)$ where x_{i-2}, x_{i-1} and x_i are approximations to a root of $f(x) = 0$. Now, a second degree curve passing through the three points is given by Lagrange's formula (see Section 3.9.1)

$$L(x) = \frac{(x - x_{i-1})(x - x_i)}{(x_{i-2} - x_{i-1})(x_{i-2} - x_i)} y_{i-2} + \frac{(x - x_{i-2})(x - x_i)}{(x_{i-1} - x_{i-2})(x_{i-1} - x_i)} y_{i-1} + \frac{(x - x_{i-2})(x - x_{i-1})}{(x_i - x_{i-2})(x_i - x_{i-1})} y_i \tag{2.38}$$

Let

$$h_i = x_i - x_{i-1},\ h_{i-1} = x_{i-1} - x_{i-2} \tag{2.39}$$

Then

$$\left.\begin{aligned} &x - x_{i-1} = x - x_i + x_i - x_{i-1} = (x - x_i) + h_i, \\ &x - x_{i-2} = x - x_i + x_i - x_{i-2} = (x - x_i) + (h_{i-1} + h_i) \\ &x_{i-2} - x_{i-1} = -h_{i-1}, \\ &x_{i-2} - x_i = -(h_{i-1} + h_i) \text{ and } \Delta_i = y_i - y_{i-1}. \end{aligned}\right\} \tag{2.40}$$

Hence

$$L(x) = \frac{(x - x_i + h_i)(x - x_i)}{h_{i-1}(h_{i-1} + h_i)} y_{i-2} + \frac{(x - x_i + h_{i-1} + h_i)(x - x_i)}{-h_{i-1} h_i} y_{i-1} + \frac{(x - x_i + h_i + h_{i-1})(x - x_i + h_i)}{h_i\,(h_{i-1} + h_i)} y_i \tag{2.41}$$

EXERCISES

2.1 Explain the bisection method for finding a real root of the equation $f(x) = 0$ and write an algorithm for its implementation with a test for relative accuracy of the approximation.

Obtain a root, correct to three decimal places, of each of the following equations using the bisection method (Problems 2.2–2.5):

2.2 $x^3 - 4x - 9 = 0$

2.3 $x^3 + x^2 - 1 = 0$

2.4 $5x \log_{10} x - 6 = 0$

2.5 $x^2 + x - \cos x = 0$

2.6 Give the sequence of steps in the regula–falsi method for determining a real root of the equation $f(x) = 0$.

Use the method of false position to find a real root, correct to three decimal places, of the following equations (Problems 2.7–2.10):

2.7 $x^3 + x^2 + x + 7 = 0$

2.8 $x^3 - x - 4 = 0$

2.9 $x = 3e^{-x}$

2.10 $x \tan x + 1 = 0$

2.11 Find the real root, which lies between 2 and 3, of the equation

$$x \log_{10} x - 1.2 = 0$$

using the methods of bisection and false–position to a tolerance of 0.5%.

2.12 Explain briefly the method of iteration to compute a real root of the equation $f(x) = 0$, stating the condition of convergence of the sequence of approximations. Give a graphical representation of the method.

Use the method of iteration to find, correct to four significant figures, a real root of each of the following equations (Problems 2.13–2.16):

2.13 $e^x = 3x$

2.14 $x = \dfrac{1}{(x+1)^2}$

2.15 $1 + x^2 = x^3$

2.16 $x - \sin x = \dfrac{1}{2}$

2.17 Establish an iteration formula to find the reciprocal of a positive number N by Newton–Raphson method. Hence find the reciprocal of 154 to four significant figures.

2.18 Explain Newton–Raphson method to compute a real root of the equation $f(x) = 0$ and find the condition of convergence. Hence, find a non-zero root of the equation $x^2 + 4\sin x = 0$.

2.19 Using Newton–Raphson method, derive a formula for finding the kth root of a positive number N and hence compute the value of $(25)^{1/4}$.

Use the Newton–Raphson method to obtain a root, correct to three decimal places, of each of the following equations (Problems 2.20–2.25):

2.20 $x^{\sin 2} - 4 = 0$

2.21 $e^x = 4x$

2.22 $x^3 - 5x + 3 = 0$

2.23 $xe^x = \cos x$

2.24 $x = \dfrac{1 + \cos x}{3}$

2.25 $\cot x = -x$

2.26 Describe a computational procedure to implement Newton–Raphson method for computing the square root of a positive number to an accuracy ε. Write a flow-chart for the same.

2.27 Compute, to four decimal places, the root between 1 and 2 of the equation

$$x^3 - 2x^2 + 3x - 5 = 0$$

by (a) Method of False Position and (b) Newton–Raphson method.

Using Ramanujan's method, find the smallest root of each of the following equations (Problems 2.28–2.30):

2.28 $x^3 - 6x^2 + 11x - 6 = 0$

2.29 $x + x^3 - 1 = 0$

2.30 $\sin x + x - 1 = 0$

2.31 Use the secant method to determine the root, between 5 and 8, of the equation $x^{2.2} = 69$. Compare your result with that obtained in Example 2.7.

2.32 Determine the real root of the equation $x = e^{-x}$, using the secant method. Compare your result with that obtained in Example 2.26.

2.33 Point out the difference between the regula–falsi and the secant methods for finding a real root of $f(x) = 0$. Apply both the methods to find a real root of the equation $x^3 - 2x - 5 = 0$ to an accuracy of 4 decimal places.

2.34 Describe briefly Muller's method and use it to find (a) the root, between 2 and 3, of the equation $x^3 - 2x - 5 = 0$ and (b) the root, between 0 and 1, of the equation $x = e^{-x} \cos x$.

Find the real roots of the following equations using Graeffe's root-squaring method (Problems 2.35–2.36):

2.35 $x^3 - 4x^2 + 5x - 2 = 0$

2.36 $x^3 - 2x^2 - 5x + 6 = 0$

2.37 Find a quadratic factor of the polynomial

$$f(x) = x^3 - 2x^2 + x - 2$$

by Bairstow's method with two approximations starting with $r_0 = -0.5$ and $s_0 = 1$.

2.38 Determine a quadratic factor, nearer to $x^2 - 1.5x + 1.5$, of the polynomial

$$f(x) = x^4 - 8x^3 + 39x^2 - 62x + 50$$

by Bairstow's method. Give the computational steps of this method.

2.39 Using the *Q-D* method, find the real roots of the equation

$$f(x) = x^3 - 6x^2 + 11x - 6 = 0$$

2.40 In the notation of Section 2.11, prove the following results:

(a) $\lim\limits_{i\to\infty} \dfrac{b_i}{b_{i-1}} = \dfrac{1}{x_1}$

(b) $\lim\limits_{i\to\infty} \dfrac{\dfrac{1}{x_1} - Q_1^{(i)}}{\left(\dfrac{x_1}{x_2}\right)^i} = \dfrac{1}{x_1}\dfrac{k_2}{k_1}\left(1 - \dfrac{x_1}{x_2}\right)$

(c) $\lim\limits_{i\to\infty} \dfrac{\dfrac{1}{x_1} - Q_1^{(i+1)}}{\left(\dfrac{x_1}{x_2}\right)^i} = \dfrac{1}{x_2}\dfrac{k_2}{k_1}\left(1 - \dfrac{x_1}{x_2}\right).$

2.41 Prove the formula

$$D_r^{(i)}\, Q_{r+1}^{(i)} = D_r^{(i+1)}\, Q_r^{(i+1)}$$

Solve the following systems by Newton–Raphson method. (Problems 2.42–2.44).

2.42 $x^2 + y^2 = 1, \quad y = x^3$

2.43 $x^2 - y^2 = 4, \quad x^2 + y^2 = 16$

2.44 $x^2 + y = 11, \quad y^2 + x = 7$

2.45 To find the smallest root of the equation $f(x) = x^3 - x - 1 = 0$ by the iteration method, $f(x) = 0$ should be rewritten as

(a) $x = x^3 - 1$ (b) $x = (x + 1)^{1/3}$

(c) $x = \dfrac{1}{x^2 - 1}$ (d) $x = \dfrac{x+1}{x^2}$

Find the correct choice in the above.

2.46 Newton–Raphson formula converges if

(a) $\left| \dfrac{f'(x) f''(x)}{[f(x)]^2} \right| < 1$ (b) $\left| \dfrac{f(x) f''(x)}{[f'(x)]^2} \right| < 1$

(c) $\left| \dfrac{f(x) f'(x)}{[f''(x)]^2} \right| < 1$ (d) None of these.

2.47 Which one of the following is not correct?

(a) Newton–Raphson method has quadratic convergence.
(b) The bisection method converges slowly.
(c) To solve $f(x) = 0$ by iteration method, the given equation is written in the form $x = \phi(x)$ where $|\phi'(x)| < 1$ in an interval containing the root.
(d) The method of regula–falsi converges faster than the secant method.

2.48 Which one of the following is correct?

(a) The bisection method has quadratic convergence.
(b) The iteration method is a self-correction method.
(c) The equation $x^3 - 2x - 5 = 0$ has two positive roots.
(d) The equation $x = \cos x$ can be solved by Graeffe's method.

Answers to Exercises

2.2 2.706 **2.3** 0.755
2.4 2.741 **2.5** 0.550
2.7 –2.105 **2.8** 1.796
2.9 1.0499 **2.10** 2.798

2.11 Successive values are (By bisection method)
2.5, 2.75, 2.625, 2.6875, 2.7187, 2.7344, 2.7422.
By false position method
2.7210, 2.7402, 2.7407, 2.7406, 2.7406

2.13 0.6190 **2.14** 0.4656
2.15 1.466 **2.16** 1.497
2.17 0.006494 **2.18** –1.9338
2.19 2.23607 **2.20** 4.5932

2.21 0.3574

2.22 With $x_0 = 0.5$, $x_1 = 0.6470$, $x_2 = 0.65656$, …

2.23 With $x_0 = 0.5$, $x_1 = 0.5180$, $x_2 = 0.5180$, …

2.24 $x_0 = 1.047$, $x_1 = 0.6224$, $x_2 = 0.6071$, $x_3 = 0.6071$.

2.25 2.798

2.27 1.8437; 1.8438

2.28 $\frac{b_1}{b_2} = 0.54546, \quad \frac{b_2}{b_3} = 0.77647,$

$\frac{b_3}{b_4} = 0.88696, \quad \frac{b_4}{b_5} = 0.94237, \ldots$

Smallest root is 1.

2.29 Convergents are:

1.0, 0.5, 0.66666, 0.75, 0.66666, 0.66666, 0.69231, 0.68421, …

2.30 Convergents are:

0.5, 0.5, 0.51064, 0.51087, 0.51097, …

Required root is 0.5110.

2.31 Root = 6.85236.

2.32 Fifth iteration value = 0.567143

2.33 Regula–falsi: 2.0945 in 7 iterations.
Secant method: 2.0946.

2.34 (a) 2.09462409 (b) 0.51752

2.35 Third approximation to the roots: 0.9168, 1.0897, 2.0019

2.36 Third approximation: 3.014443, 1.991425, 0.999494.

2.37 $x^2 - 0.0165x + 0.9394$ (Exact factor is $x^2 + 1$).

2.38 $x^2 - 1.9485x + 1.942982$ (Exact factor is $x^2 - 2x + 2$).

2.39 $Q_1 \approx 3.13145$, $Q_2 \approx 1.898$, $Q_3 \approx 0.9706$.

2.42 $x_1 = 0.8261$, $y_1 = 0.5636$.

2.43 $x_0 = 3$, $y_0 = 2$, $x_1 = 3.1667$, $y_1 = 2.5$

2.44 $x = 3.584$, $y = -1.848$

2.45 (b)

2.46 (b)

2.47 (d)

2.48 (a)